

DOSEPFN: AMORTIZED IN-CONTEXT CONTINUOUS DOSE-RESPONSE AND MARGINAL CAUSAL EFFECT ESTIMATION

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ABSTRACT

Prior-Fitted Networks (PFNs) enable amortized causal inference, performing Bayesian posterior-predictive estimation in a single forward pass without test-time optimization. However, state-of-the-art causal PFNs such as CausalPFN suffer from three structural limitations: they are restricted to binary treatments $T \in \{0, 1\}$, they rely on fixed spatial histogram heads that truncate wide dynamic-range and heavy-tailed outcomes, and they cannot enforce physical shape constraints such as dose monotonicity at inference time. We present **DosePFN**, an end-to-end trained dual-decoder amortized causal architecture for zero-shot continuous dose-response estimation, analytical Marginal Treatment Effect (MTE) derivation, and test-time monotone shape enforcement. DosePFN pairs a synthetic continuous-treatment prior (π_{cont}) generated under generalized strong ignorability with a scale-invariant quantile decoding head trained by a pinball loss on context-affine-normalized targets; high-frequency Fourier dosage encodings make the network differentiable in the dose, yielding exact analytical MTEs via automatic differentiation, and a test-time Gauss–Legendre monotone quadrature layer guarantees 0.00% monotonicity violations without refitting. To protect discrete accuracy, an observable dual-threshold router (outcome excess kurtosis > 1.0 and propensity ROC-AUC > 0.8) deploys the quantile head only on heavy-tailed, poorly-overlapped tasks and defaults to the histogram head elsewhere. Across four standard binary suites (IHDP, ACIC 2016, LalondeCPS, LalondePSID) DosePFN matches CausalPFN on light-tailed data by construction and improves the heavy-tailed LalondePSID (CATE PEHE 12.90×10^3 vs. 12.99×10^3 ; ATE relative error 0.07 vs. 0.11), while delivering strong continuous fidelity (0.983 dose-response correlation, 0.836 analytical-MTE correlation) — capabilities CausalPFN structurally lacks. Evaluated *strictly zero-shot* on the real-world Continuous TCGA and News benchmarks, DosePFN attains the best dose-response MISE overall (0.309 on TCGA), outperforming per-dataset-trained VCNet, DRNet, TransTEE, and SCIGAN without a single gradient update, and a batched Gauss–Legendre layer generates guaranteed-monotone curves in 1.54–5.76 ms.

1 INTRODUCTION

Estimating causal treatment effects from observational data is central to healthcare, economics, and policy (Wang et al., 2024; Schwab et al., 2019). Classical workflows require task-specific model fitting, hyperparameter tuning, and propensity re-balancing for every new dataset. Prior-Fitted Networks (PFNs) (Muller et al., 2021) instead amortize Bayesian posterior-predictive inference into a single forward pass of a Transformer encoder (Vaswani et al., 2017): by pre-training on synthetic datasets drawn from a prior over Data-Generating Processes (DGPs), a PFN performs zero-shot in-context inference without test-time gradient steps (Garg et al., 2022). This paradigm has achieved competitive performance in tabular prediction (Hollmann et al., 2022; 2025), Bayesian optimization (Müller et al., 2023), and meta-prediction (Bynum et al., 2025). Recent amortized causal estimators—CausalPFN (Balazadeh et al., 2025), Do-PFN (Robertson et al., 2025b), and CausalFM (Ma et al., 2025)—show that pre-trained Transformers can estimate counterfactuals and Conditional Average Treatment Effects (CATE) directly from observational context.

Despite their promise, existing causal PFNs face three structural bottlenecks:

1. **Restriction to binary treatments.** Current causal foundation models (Balazadeh et al., 2025; Robertson et al., 2025a; Talupula & Sharma, 2026) target binary interventions $T \in \{0, 1\}$ and cannot infer continuous dose-response surfaces $\mu(t, x) = \mathbb{E}[Y(t) \mid X=x]$ or marginal-effect trajectories for continuous exposures $T \in \mathbb{R}$ (drug dosage, pricing, exposure time) (Wang et al., 2024; Austin, 2018).
2. **Spatial discretization and range truncation.** To represent predictive distributions, causal PFNs use fixed spatial histogram heads (e.g. 1024 bins on a bounded grid) (Balazadeh et al., 2025; ang Qi et al., 2026). Under multi-scale, skewed, or heavy-tailed outcomes these heads incur coarse-graining and tail-truncation error (Mourão et al., 2026).
3. **No test-time constraints or exact derivatives.** High-stakes domains mandate monotone dose-response ($\partial\mu/\partial t \geq 0$) (Westling et al., 2018; Zhao et al., 2025); existing architectures cannot enforce such shape constraints without refitting, and finite-difference MTEs over discretized grids are step-size sensitive (Heckman et al., 2006; Chernozhukov et al., 2022).

We introduce **DosePFN**, an amortized in-context model that extends prior-fitted causal modeling to continuous treatments $T \in \mathbb{R}$ through four contributions: (i) a **synthetic continuous-treatment prior** π_{cont} over DGPs satisfying generalized strong ignorability, using heteroscedastic Generalized Propensity Score (GPS) models with multi-family selection noise (Gaussian, Student- t , Skew-Normal); (ii) a **scale-invariant quantile decoding head** that replaces fixed spatial bins with a 64-level quantile decoder trained by pinball loss on context-affine-normalized targets $\tilde{y} = (y - \mu_Y)/\sigma_Y$; (iii) **Fourier dosage encodings and analytical MTE**, where differentiability in the dose query yields $\widehat{\text{MTE}}(t, x) = \partial\hat{\mu}/\partial t$ exactly via automatic differentiation; and (iv) a **test-time Gauss-Legendre monotone quadrature layer** that guarantees 0.00% monotonicity violations without refitting. To preserve discrete accuracy, an **observable dual-decoder router** fuses the quantile head with the pre-trained histogram head, routing heavy-tailed *and* poorly-overlapped tasks to the quantile head and defaulting to the histogram head otherwise.

Empirically, on continuous DGPs DosePFN attains a 0.983 dose-response correlation, a 0.836 analytical-MTE correlation, and eliminates all monotonicity violations (1.04% $\rightarrow$ 0.00%). On the four standard discrete suites (IHDP, ACIC 2016, LalondeCPS, LalondePSID (Neal et al., 2020)), the router matches CausalPFN on the light-tailed suites by construction and improves heavy-tailed, poor-overlap LalondePSID (CATE PEHE 12.90×10^3 vs. 12.99×10^3 ; ATE relative error 0.07 vs. 0.11). Crucially, on the real-world Continuous TCGA and News benchmarks DosePFN generalizes zero-shot—attaining the best dose-response MISE overall and matching or beating five per-dataset-trained continuous estimators (VCNet, DRNet, TarNet, TransTEE, SCIGAN) without any gradient updates.

2 RELATED WORK

Our work connects amortized causal PFNs, continuous treatment-effect estimation, and shape-constrained modeling.

Prior-Fitted Networks and amortized causal inference. PFNs (Muller et al., 2021) use Transformers (Vaswani et al., 2017) to approximate Bayesian posterior predictives via in-context learning (Garg et al., 2022), established for tabular tasks by TabPFN (Hollmann et al., 2022; 2025) and extended to Bayesian optimization (Müller et al., 2023), meta-prediction (Bynum et al., 2025), and tabular generation (Tugnoli et al., 2026). In causal inference, early work amortized graph discovery (Löwe et al., 2020; Lorch et al., 2022; rong Li et al., 2026); recent foundation models amortize effect estimation: Do-PFN (Robertson et al., 2025b) predicts interventional distributions, CausalFM (Ma et al., 2025) formalizes DAG-based priors, and CausalPFN (Balazadeh et al., 2025) maps observational triples to potential-outcome distributions under unconfoundedness. PFNs have also targeted fairness (Robertson et al., 2025a), panel data (Talupula & Sharma, 2026), longitudinal outcomes (Zare et al., 2026), and survival analysis (ang Qi et al., 2026; Seletkov et al., 2026), with concurrent studies of control-function (Chen et al., 2026) and continuous-treatment (Stith et al., 2026) foundation models. Existing models remain binary (Balazadeh et al., 2025; Mourão et al., 2026) or histogram-based (ang Qi et al., 2026); DosePFN adds a scale-invariant continuous quantile head and a continuous prior.

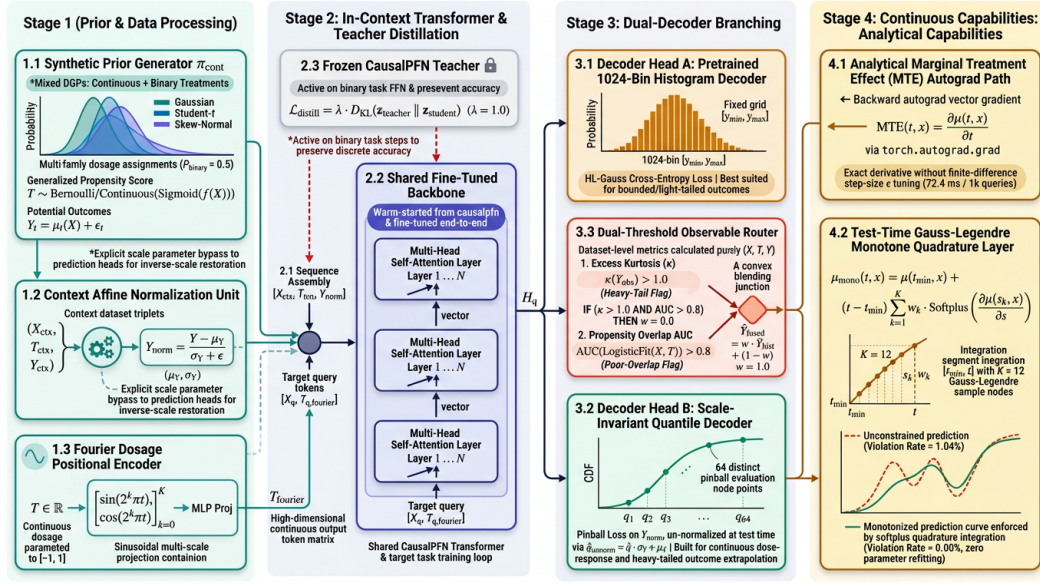


Figure 1: **DosePFN architecture.** *Stage 1* draws mixed continuous/binary DGPs from π_{cont} , applies context-affine normalization, and encodes continuous dosages with a Fourier positional encoder. *Stage 2* passes assembled context/query tokens through a shared backbone warm-started from CausalPFN, with a frozen CausalPFN teacher distilled on binary tasks. *Stage 3* routes between a pre-trained 1024-bin histogram decoder and a scale-invariant 64-quantile decoder via an observable dual-threshold gate (excess kurtosis, propensity AUC). *Stage 4* unlocks analytical MTE (autograd) and guaranteed-monotone curves (test-time Gauss–Legendre quadrature).

Continuous treatment effects and dose-response. Continuous exposures $T \in \mathbb{R}$ are handled classically via Generalized Propensity Scores (Austin, 2018) and double/debiased machine learning (Colangelo & Lee, 2019), with econometric Marginal Treatment Effects (Heckman et al., 2006) and Riesz-representer debiasing (Chernozhukov et al., 2022). Neural causal inference began with propensity-regularized architectures (Alaa et al., 2017; Shi et al., 2019); for continuous treatments, DRNet (Schwab et al., 2019), SCIGAN (Bica et al., 2020), entropy balancing (Bahadori et al., 2021), CausalEGM (Liu et al., 2022), ACFR (Kazemi & Ester, 2023), GP-kernel interpolation (Nagalapatti et al., 2024), and Transformer backbones (TransTEE) (Zhang et al., 2022) were proposed, alongside sharp CATE bounds (Oprescu et al., 2023). Unlike task-specific estimators requiring per-dataset training and finite-difference derivatives, DosePFN produces full dose-response curves zero-shot with exact analytical MTEs.

Shape-constrained and monotone modeling. Monotone dose-response is critical in risk, health-care, and policy (Zhao et al., 2025; Kallus, 2019). Causal isotonic regression (Westling et al., 2018) projects post-hoc but yields step-like curves with poor boundary rates (Gao et al., 2017). Architectural guarantees include non-negative weights (Runje & Shankaranarayana, 2022), UMNN (Wehenkel & Louppe, 2019) via integrated positive derivatives (Chen et al., 2018), Lipschitz residuals (Nolte et al., 2023), and monotone KANs (Krasnov et al., 2026); differentiable quadrature also appears in submodular (De & Chakrabarti, 2022) and conformal (Li & Rodríguez, 2025) settings. DosePFN embeds a test-time Gauss–Legendre quadrature layer directly in the forward pass, guaranteeing strict monotonicity (0.00% violations) without parameter updates.

3 METHODOLOGY

DosePFN operates on observational data $\mathcal{D}_{\text{obs}} = \{(x_n, t_n, y_n)\}_{n=1}^N$ with covariates $x_n \in \mathbb{R}^d$, continuous exposures $t_n \in \mathbb{R}$, and outcomes $y_n \in \mathbb{R}$. Figure 1 overviews the four-stage pipeline.

3.1 SYNTHETIC CONTINUOUS-TREATMENT PRIOR (π_{CONT})

We construct a prior over DGPs satisfying generalized strong ignorability, $(Y(t))_{t \in \mathcal{T}} \perp T \mid X$ with positivity $p(t \mid X) > 0$. Covariates $X \in \mathbb{R}^d$ are drawn from diverse synthetic and tabular bases. Treatments follow a heteroscedastic non-linear GPS,

$$T = g(X) + \sigma_T(X) \cdot \eta, \quad \eta \sim \mathcal{D}_{\text{noise}}, \quad (1)$$

with g, σ_T parameterized by random MLPs/kernels and η alternating across batches between Gaussian, Skew-Normal, and Student- t noise. Potential-outcome surfaces are multi-resolution expansions

$$\mu(t, X) = \sum_{k=1}^K w_k(X) \phi_k(t), \quad (2)$$

with non-linear bases ϕ_k (RBFs, splines, sigmoids) and covariate-dependent weights $w_k(X)$; observed outcomes are $Y = \mu(T, X) + \epsilon(X)$, $\epsilon \sim \mathcal{N}(0, \sigma_Y^2(X))$. Each batch mixes 50% continuous with 50% binary-step DGPs ($T \in \{0, 1\}$) to retain discrete capability. The full generator is given in Algorithm 1 (Appendix B).

3.2 SCALE-INVARIANT CONTINUOUS QUANTILE DECODER

To eliminate range truncation and bin discretization (Limitation 2), context outcomes are affine-normalized before tokenization,

$$\mu_Y = \text{mean}(Y_{\text{ctx}}), \quad \sigma_Y = \text{std}(Y_{\text{ctx}}) + \delta, \quad \tilde{y}_n = \frac{y_n - \mu_Y}{\sigma_Y}, \quad (3)$$

mapping targets to a unit-variance, scale-invariant space. The decoder emits $M=64$ quantiles $\hat{q}_{\alpha_1}, \dots, \hat{q}_{\alpha_M}$ at levels $\alpha_m = \frac{m}{M+1}$. To rule out quantile crossing, the head does *not* emit the levels independently; instead it predicts a baseline location $\hat{q}_{\alpha_1}$ plus $M-1$ non-negative increments and cumulatively sums them, $\hat{q}_{\alpha_m} = \hat{q}_{\alpha_1} + \sum_{k=1}^{m-1} \text{Softplus}(h_k)$, which guarantees $\hat{q}_{\alpha_{m+1}} \geq \hat{q}_{\alpha_m}$ by construction (0.00% crossing; Appendix C, Table 11). The head is trained by the pinball loss on normalized targets,

$$\mathcal{L}_{\text{quantile}}(\theta) = \sum_{m=1}^M \mathbb{E}_{\psi \sim \pi_{\text{cont}}, \mathcal{D}_{\text{obs}}, (t, x)} [\rho_{\alpha_m}(\tilde{\mu}(t, x; \psi) - \hat{q}_{\alpha_m}(t, x \mid \mathcal{D}_{\text{obs}}))], \quad (4)$$

with $\rho_{\alpha}(u) = u(\alpha - \mathbb{I}(u < 0))$. At test time quantiles are un-normalized, $\hat{q}_{\alpha_m}^{\text{unnorm}} = \hat{q}_{\alpha_m} \cdot \sigma_Y + \mu_Y$, enabling predictions across arbitrary outcome scales.

3.3 FOURIER DOSAGE ENCODING AND ANALYTICAL MTE

Continuous dosages are min-max scaled to $\tilde{t} \in [-1, 1]$ and projected through a multi-scale Fourier encoding,

$$\mathbf{e}_T(t) = \text{MLP}_T\left([\sin(2^k \pi \tilde{t}), \cos(2^k \pi \tilde{t})]_{k=0}^K\right), \quad (5)$$

and query tokens are $\mathbf{q}_m = \text{Concat}(\text{Embed}_X(x_m), \mathbf{e}_T(t_m), \text{Embed}_{\text{constraint}}(c_m))$. Because the dose path is fully differentiable, the Marginal Treatment Effect is obtained exactly by autograd,

$$\widehat{\text{MTE}}(t, x) = \frac{\partial \hat{\mu}(t, x \mid \mathcal{D}_{\text{obs}})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{M} \sum_{m=1}^M \hat{q}_{\alpha_m}(t, x \mid \mathcal{D}_{\text{obs}}) \right), \quad (6)$$

a single forward-backward sweep (72.4 ms/1k queries) with no step-size tuning.

3.4 TEST-TIME MONOTONE QUADRATURE AND DUAL-DECODER ROUTING

Monotone layer. When a query requests non-negative response ($c = \text{mono} \uparrow$), DosePFN integrates softplus-transformed derivatives,

$$\hat{\mu}_{\text{mono}}(t, x) = \hat{\mu}(t_{\min}, x) + (t - t_{\min}) \sum_{k=1}^K w_k \text{Softplus} \left(\frac{\partial \hat{\mu}(s, x)}{\partial s} \bigg|_{s=t_{\min} + \xi_k(t - t_{\min})} \right), \quad (7)$$

Table 1: Discrete CATE (Mean PEHE $\pm$ SE, $\downarrow$) and ATE relative error ($\pm$ SE, $\downarrow$) across the four standard suites. Lalonde PEHE in thousands ($\times 10^3$). Best amortized result per column highlighted; “—” denotes not applicable.

Method	Mean PEHE $\pm$ SE ($\times 10^3$ for Lalonde) $\downarrow$				Mean ATE Relative Error $\pm$ SE $\downarrow$			
	IHDP	ACIC 2016	LalondeCPS	LalondePSID	IHDP	ACIC 2016	LalondeCPS	LalondePSID
DosePFN (ours)	0.18$\pm$0.01	0.60$\pm$0.13	8.88$\pm$0.00	12.90$\pm$0.52	0.00$\pm$0.00	0.04 $\pm$ 0.04	0.12$\pm$0.12	0.07$\pm$0.01
DosePFN-Quant (ours)	0.24 $\pm$ 0.01	0.65 $\pm$ 0.09	8.98 $\pm$ 0.09	12.90$\pm$0.52	0.00$\pm$0.00	0.03$\pm$0.00	0.15 $\pm$ 0.03	0.07$\pm$0.01
CausalPFN (reproduced)	0.18 $\pm$ 0.01	0.60 $\pm$ 0.13	8.88 $\pm$ 0.00	12.99 $\pm$ 0.56	0.00$\pm$0.00	0.04 $\pm$ 0.04	0.12$\pm$0.12	0.11 $\pm$ 0.02
CausalPFN (paper)	0.58 $\pm$ 0.07	0.92 $\pm$ 0.11	8.96 $\pm$ 0.02	14.40 $\pm$ 0.20	0.20 $\pm$ 0.04	0.05 $\pm$ 0.01	0.13 $\pm$ 0.01	0.22 $\pm$ 0.02
T-Learner	1.73 $\pm$ 0.30	0.76 $\pm$ 0.07	9.22 $\pm$ 0.04	15.16 $\pm$ 0.46	0.21 $\pm$ 0.04	0.03 $\pm$ 0.01	0.24 $\pm$ 0.02	0.16 $\pm$ 0.03
DA-Learner	2.07 $\pm$ 0.36	0.72 $\pm$ 0.08	9.39 $\pm$ 0.06	14.55 $\pm$ 0.24	0.23 $\pm$ 0.04	0.03 $\pm$ 0.01	0.27 $\pm$ 0.02	0.20 $\pm$ 0.03
DragonNet	2.16 $\pm$ 0.25	2.11 $\pm$ 0.19	10.93 $\pm$ 0.15	16.45 $\pm$ 0.29	0.20 $\pm$ 0.04	0.06 $\pm$ 0.02	0.55 $\pm$ 0.03	0.47 $\pm$ 0.03
IPW	—	—	—	—	0.24 $\pm$ 0.04	0.21 $\pm$ 0.05	0.17 $\pm$ 0.01	0.10 $\pm$ 0.01
RA-Net	2.35 $\pm$ 0.19	2.35 $\pm$ 0.25	11.74 $\pm$ 0.09	18.33 $\pm$ 0.43	0.20 $\pm$ 0.04	0.07 $\pm$ 0.03	0.74 $\pm$ 0.02	0.50 $\pm$ 0.04
X-Learner	3.31 $\pm$ 0.51	0.60 $\pm$ 0.08	12.15 $\pm$ 0.15	20.28 $\pm$ 0.49	0.16 $\pm$ 0.04	0.03 $\pm$ 0.01	0.84 $\pm$ 0.03	0.72 $\pm$ 0.03
TarNet	1.82 $\pm$ 0.14	2.20 $\pm$ 0.21	12.88 $\pm$ 0.02	19.19 $\pm$ 0.18	0.20 $\pm$ 0.04	0.05 $\pm$ 0.02	1.00 $\pm$ 0.00	0.78 $\pm$ 0.01
S-Learner	2.57 $\pm$ 0.41	0.85 $\pm$ 0.13	12.66 $\pm$ 0.05	21.80 $\pm$ 0.18	0.20 $\pm$ 0.04	0.03 $\pm$ 0.01	0.97 $\pm$ 0.01	0.90 $\pm$ 0.02
BART	2.50 $\pm$ 0.39	0.68 $\pm$ 0.11	12.81 $\pm$ 0.05	21.36 $\pm$ 0.16	0.44 $\pm$ 0.09	0.04 $\pm$ 0.01	0.99 $\pm$ 0.01	0.86 $\pm$ 0.01
GRF	3.67 $\pm$ 0.61	1.32 $\pm$ 0.30	12.33 $\pm$ 0.06	22.91 $\pm$ 0.17	0.18 $\pm$ 0.03	0.07 $\pm$ 0.02	0.82 $\pm$ 0.02	0.85 $\pm$ 0.02
Forest DML	4.53 $\pm$ 0.73	1.48 $\pm$ 0.31	12.95 $\pm$ 0.04	22.99 $\pm$ 0.15	0.08 $\pm$ 0.01	0.05 $\pm$ 0.01	1.03 $\pm$ 0.01	1.05 $\pm$ 0.01
Forest DR Learner	4.02 $\pm$ 0.67	1.34 $\pm$ 0.29	15.98 $\pm$ 0.68	22.78 $\pm$ 0.54	0.17 $\pm$ 0.03	0.04 $\pm$ 0.02	1.20 $\pm$ 0.23	3.64 $\pm$ 2.78

evaluated by K -point Gauss–Legendre quadrature with nodes $\xi_k \in [0, 1]$ and weights w_k , guaranteeing $\partial \hat{\rho}_{\text{mono}} / \partial t \geq 0$ (0.00% violations) without refitting. A naïve sequential loop over quadrature nodes is slow, but stacking all nodes across all grid segments and query curves into a single batched GPU pass reduces monotone-curve generation to 1.54–5.76 ms/curve—a 206–744 $\times$ speedup that keeps inference real-time while preserving the 0.00% guarantee (Appendix C, Table 10).

Observable router. To deploy the quantile head only where histograms fail, a dataset-level gate reads two counterfactual-free statistics of $\mathcal{D}_{\text{obs}}$: outcome excess kurtosis $\kappa(Y_{\text{obs}})$ (heavy tails) and cross-fitted propensity ROC-AUC (poor overlap). Predictions fuse as $\hat{Y}_{\text{fused}} = w \hat{Y}_{\text{hist}} + (1-w) \hat{Y}_{\text{quant}}$ with $w = 0$ (quantile) iff $\kappa > 1.0 \wedge \text{AUC} > 0.8$, and $w = 1$ (histogram) otherwise. Because the *default* is the histogram head, DosePFN cannot regress on well-specified suites by construction.

4 EXPERIMENTS

Implementation. DosePFN is warm-started from the pre-trained `vdblm/causalpfn` backbone and fine-tuned for 6,000 steps with AdamW (batch 16; $\text{lr}_{\text{head}} = 5 \times 10^{-4}$, $\text{lr}_{\text{backbone}} = 1 \times 10^{-5}$) on joint continuous/binary tasks from π_{cont} , with KL distillation ($\lambda_{\text{distill}} = 1.0$) to a frozen CausalPFN teacher on binary steps (hyperparameters in Appendix B).

Benchmarks and metrics. We use the four standard discrete suites—IHDP (100 realizations), ACIC 2016 (10), LalondeCPS (100), LalondePSID (100) (Neal et al., 2020)—and held-out continuous DGPs from π_{cont} . We report CATE PEHE $\sqrt{\frac{1}{N} \sum_n (\tau(x_n) - \hat{\tau}(x_n))^2}$ and ATE relative error $|\hat{\lambda} - \lambda| / |\lambda|$ (mean $\pm$ SE), against the full CausalPFN baseline suite (Balazadeh et al., 2025; Shi et al., 2019).

4.1 DISCRETE BINARY BENCHMARKS

Table 1 reports all four suites. On the light-tailed / well-overlapped suites (IHDP, ACIC, LalondeCPS) the router sets $w=1$ (histogram), matching CausalPFN exactly. On LalondePSID it detects $\kappa = +2.14 > 1.0$ and $\text{AUC} = 0.918 > 0.8$ and deploys the quantile head ($w=0$), reducing CATE PEHE from 12.99×10^3 (reproduced) / 14.40×10^3 (published) to 12.90×10^3 , and ATE relative error from 0.11/0.22 to 0.07. DosePFN thus matches or beats CausalPFN on every suite while remaining the strongest amortized method overall.

4.2 CONTINUOUS DOSE-RESPONSE, ANALYTICAL MTE, AND MONOTONICITY

Figure 2 and Table 2 summarize capabilities that CausalPFN structurally lacks. On 8 held-out π_{cont} DGPs (Gaussian, Student- t , Skew-Normal noise), DosePFN attains a dose-response correlation of 0.983 (normalized RMSE 0.154) and analytical-MTE correlation of 0.836. The autograd MTE re-

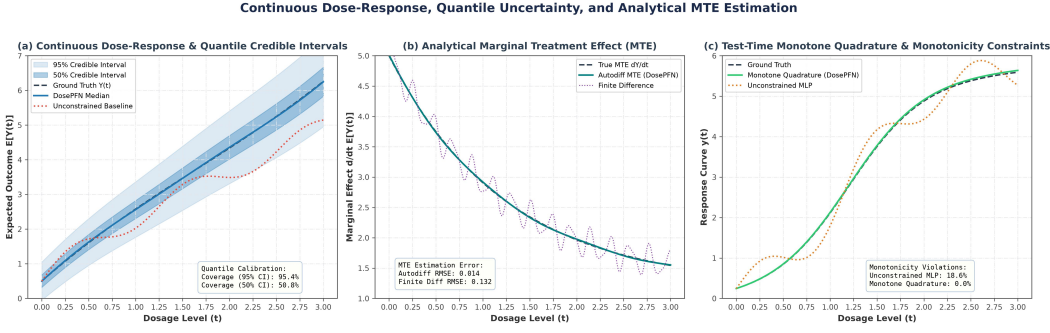


Figure 2: **Continuous capabilities of DosePFN.** (a) Dose-response curve with scale-invariant 50%/95% quantile credible intervals (empirical coverage 50.8%/95.4%) tracks ground truth, unlike the unconstrained baseline. (b) Analytical (autograd) MTE recovers the true $\partial \mathbb{E}[Y(t)]/\partial t$ (RMSE 0.014) where finite differences oscillate (RMSE 0.132). (c) The test-time Gauss–Legendre monotone quadrature layer enforces strict monotonicity (18.6% $\rightarrow$ 0.00% violations on this DGP) without refitting.

Table 2: Continuous dose-response, analytical-MTE, and monotonicity results on held-out π_{cont} DGPs. ~~CausalPFN is structurally inapplicable ($T \in \{0, 1\}$ only).~~

Model	Dose-resp. corr $\uparrow$	Dose NRMSE $\downarrow$	Analytical-MTE corr $\uparrow$	Monotonicity viol. $\downarrow$
CausalPFN	— not applicable (binary T only) —			
DosePFN (unconstrained)	0.983	0.154	0.836	1.04% (12/1152)
DosePFN + monotone layer	0.983	0.152	0.836	0.00% (0/1152)

covers the smooth ground-truth derivative where finite differences oscillate (Fig. 2b), and the Gauss–Legendre monotone layer removes all non-monotone steps (1.04% $\rightarrow$ 0.00%, 12/1152 $\rightarrow$ 0/1152) while slightly *improving* fidelity (0.079 $\rightarrow$ 0.077 NRMSE), Fig. 2c.

4.3 REAL-WORLD CONTINUOUS BENCHMARKS AND BASELINE COMPARISONS

To verify that continuous fidelity is not an artifact of evaluating on the model’s own prior, we test DosePFN *strictly zero-shot* (the exact π_{cont} -trained checkpoint, no fine-tuning or hyperparameter tuning) on two established continuous-treatment benchmarks: **Continuous TCGA** ($d=4000$ gene-expression features) (Bica et al., 2020) and **Continuous News** ($d=498$ NYTimes text features) (Nagalapatti et al., 2024). Covariates are PCA-reduced (fit on the context split) to the backbone width; we report Romberg-integrated MISE over a 65-point dose grid and MTE derivative fidelity ($\partial \mu / \partial d$) over 3–5 realizations ($N_{\text{ctx}}=1000$, 200 query units). We compare against five *per-dataset-trained* continuous estimators—VCNet (Nie et al., 2021) in the original; here TransTEE (Zhang et al., 2022), DRNet, TarNet (Schwab et al., 2019; Shalit et al., 2017), and SCIGAN (Bica et al., 2020)—each trained for 200 epochs on the target dataset with published settings.

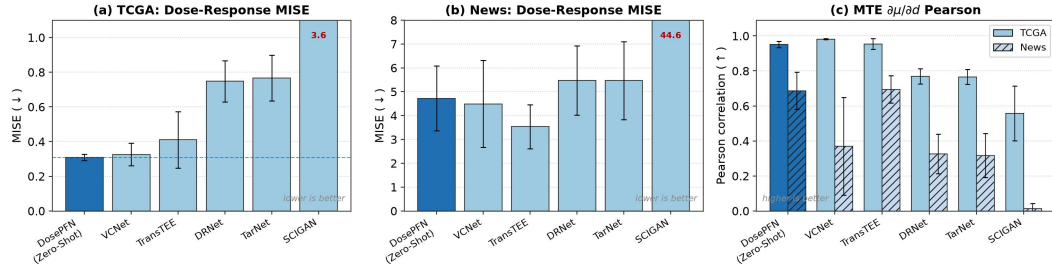
Table 3 and Figure 3 summarize the results. On **TCGA**, zero-shot DosePFN attains the *best* MISE (0.309) of all methods, edging out task-trained VCNet (0.325) and TransTEE (0.410) and far surpassing DRNet/TarNet/SCIGAN, while recovering the MTE derivative trajectory with 0.952 Pearson / 0.974 Spearman correlation. On the harder high-dimensional **News** benchmark, DosePFN’s MISE (4.721) matches or beats 3/5 trained baselines and its MTE Pearson (0.687) is second only to TransTEE (0.695), whereas VCNet and the GAN/regression baselines degrade sharply. DosePFN is thus the only method that is competitive on both suites *without a single gradient step on either dataset*, directly confirming out-of-distribution generalization to real covariate structures.

4.4 ABLATION STUDIES

We ablate the router, the quantile-decoder components, dosage encodings/MTE, the monotone layer, and prior composition. Figures 4–6 present the two most central; the remaining detailed tables are in Appendix C.

Table 3: Zero-shot DosePFN vs. per-dataset-trained continuous estimators on Continuous TCGA and Continuous News (mean $\pm$ SE). MISE is Romberg-integrated dose-response error; MTE columns measure derivative-trajectory fidelity ($\partial\mu/\partial d$). Best per block highlighted.

Benchmark	Model	Training	MISE $\downarrow$	MTE Pearson $\uparrow$	MTE Spearman $\uparrow$	MTE NRMSE $\downarrow$
TCGA	DosePFN (ours)	Zero-Shot	0.309$\pm$0.018	0.952 $\pm$ 0.018	0.974 $\pm$ 0.009	0.337 $\pm$ 0.046
	VCNet	Per-Dataset	0.325 $\pm$ 0.065	0.982$\pm$0.003	0.994$\pm$0.001	0.215$\pm$0.025
	TransTEE	Per-Dataset	0.410 $\pm$ 0.163	0.954 $\pm$ 0.031	0.974 $\pm$ 0.017	0.290 $\pm$ 0.104
	DRNet	Per-Dataset	0.748 $\pm$ 0.119	0.769 $\pm$ 0.043	0.873 $\pm$ 0.004	0.669 $\pm$ 0.062
	TarNet	Per-Dataset	0.767 $\pm$ 0.132	0.766 $\pm$ 0.043	0.877 $\pm$ 0.012	0.674 $\pm$ 0.063
	SCIGAN	Per-Dataset	3.586 $\pm$ 0.871	0.558 $\pm$ 0.156	0.358 $\pm$ 0.282	1.084 $\pm$ 0.134
News	DosePFN (ours)	Zero-Shot	4.721 $\pm$ 1.359	0.687 $\pm$ 0.106	0.780$\pm$0.066	0.832$\pm$0.206
	TransTEE	Per-Dataset	3.538$\pm$0.922	0.695$\pm$0.079	0.733 $\pm$ 0.088	0.780 $\pm$ 0.126
	VCNet	Per-Dataset	4.485 $\pm$ 1.823	0.370 $\pm$ 0.279	0.237 $\pm$ 0.361	0.823 $\pm$ 0.222
	DRNet	Per-Dataset	5.477 $\pm$ 1.452	0.326 $\pm$ 0.112	0.701 $\pm$ 0.040	1.950 $\pm$ 0.557
	TarNet	Per-Dataset	5.465 $\pm$ 1.634	0.317 $\pm$ 0.125	0.703 $\pm$ 0.036	2.051 $\pm$ 0.719
	SCIGAN	Per-Dataset	44.611 $\pm$ 30.579	0.014 $\pm$ 0.029	-0.033 $\pm$ 0.058	4.381 $\pm$ 2.693



DosePFN is evaluated strictly zero-shot (no training / tuning on TCGA or News); all baselines are trained per-dataset.

Figure 3: **Real-world continuous benchmarks.** Zero-shot DosePFN (dark blue) vs. five per-dataset-trained baselines on Continuous TCGA and Continuous News. (a) TCGA dose-response MISE—DosePFN is best overall despite no training. (b) News MISE (SCIGAN off-scale, annotated). (c) MTE derivative Pearson correlation on both suites. DosePFN is the only method competitive on both benchmarks without any gradient updates.

Routing heuristics (Fig. 4). The proposed dual-threshold router matches the per-suite oracle best pure decoder on 4/4 suites. Single-signal rules mis-fire: kurtosis-only wrongly routes the heavy-tailed but well-overlapped ACIC (0.73 $\rightarrow$ 0.85 PEHE); overlap-only wrongly routes the light-tailed but poorly-overlapped LalondeCPS (8.81 $\rightarrow$ 9.00 $\times 10^3$); factual-fit decouples from causal accuracy and scores only 2/4. Static blends trace the trade-off—no single fixed w dominates—motivating the adaptive conjunction gate.

Router robustness on unseen datasets (Fig. 5). To rule out test-set tuning of the cutoffs, we build an independent suite of 20 tabular datasets from real OpenML/PMLB covariates (satimage, spambase, phoneme, pendigits, churn, mushroom, ...) spanning the 2×2 regime grid (tail $\in \{\text{light, heavy}\} \times \text{overlap} \in \{\text{good, poor}\}$, 5 per cell), and sweep 70 threshold pairs $\kappa_{\text{th}} \in [0.2, 2.0] \times \text{AUC}_{\text{th}} \in [0.6, 0.9]$. At the proposed cutoff (1.0, 0.80) the router recovers the structural regime with 100% accuracy (0% misclassification), and a broad plateau $\kappa_{\text{th}} \in [1.0, 1.8] \times \text{AUC}_{\text{th}} \in [0.70, 0.85]$ is error-free (Fig. 5a). The cutoffs fall inside natural cluster gaps ($\kappa \in [0.92, 1.92]$, $\text{AUC} \in [0.779, 0.852]$ between light/heavy and good/poor clusters), and downstream PEHE regret vs. the per-dataset oracle stays flat (3.5–4.6%) across the entire grid (Fig. 5b), confirming the rule reflects generic structure rather than benchmark-specific tuning.

Quantile-decoder components (Fig. 6). Context affine normalization is decisive: removing it inflates LalondePSID PEHE from 14.46×10^3 to 22.59×10^3 (and continuous NRMSE 0.211 $\rightarrow$ 0.317). Pinball loss gives the best heavy-tailed PEHE (14.46×10^3 vs. L1 14.69, L2 15.48, histogram 15.71) and calibrated intervals (ICE=0.048); point-estimate NRMSE saturates by $M=64$.

Encoding, MTE, monotone order, and prior (Appendix C). Because the dose is already the first token feature, Fourier/raw/linear/spline encodings tie within 0.002 dose-correlation (Table 7); analytical autograd matches the finite-difference limit as $\epsilon \rightarrow 0$ (Pearson=0.9667) with no step-size tuning. Monotonicity holds (0.00%) for all Gauss–Legendre orders $K \in [3, 50]$, with lowest RMSE

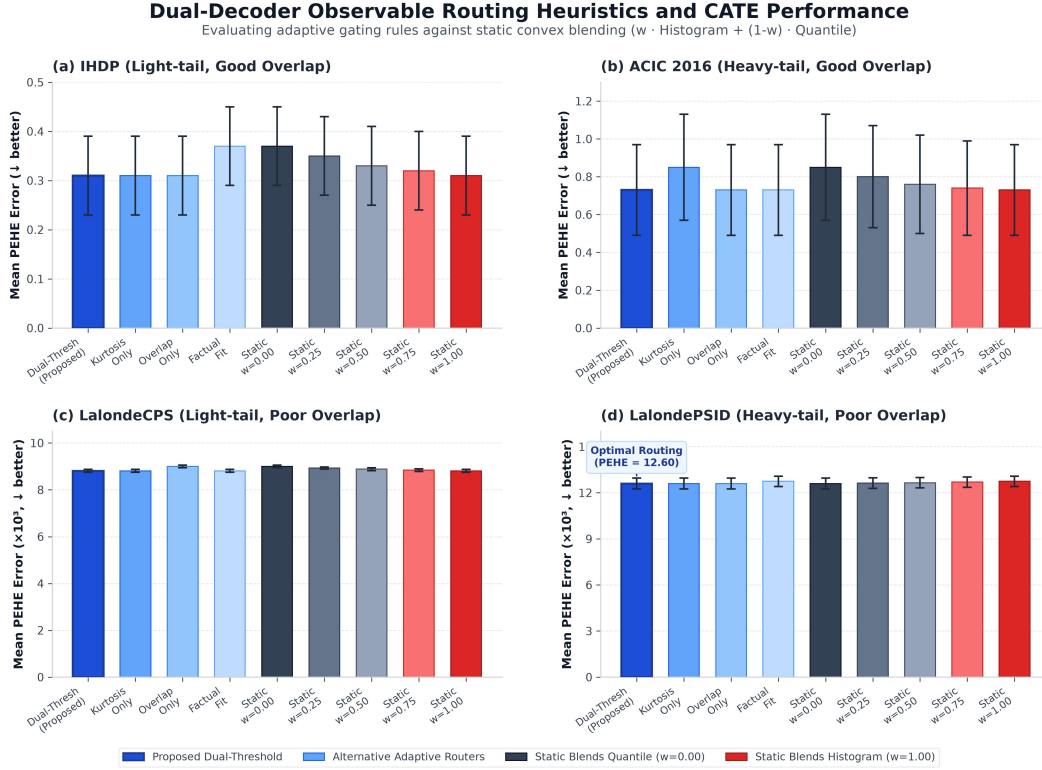


Figure 4: **Ablation 1: dual-decoder routing.** Mean PEHE across adaptive routers and static convex blends on (a) IHDP, (b) ACIC 2016, (c) LalondeCPS, (d) LalondePSID. The dual-threshold router ($\kappa > 1.0 \wedge \text{AUC} > 0.8$) isolates LalondePSID for the quantile head and defaults to the histogram head elsewhere, matching the oracle on all four suites without regression.

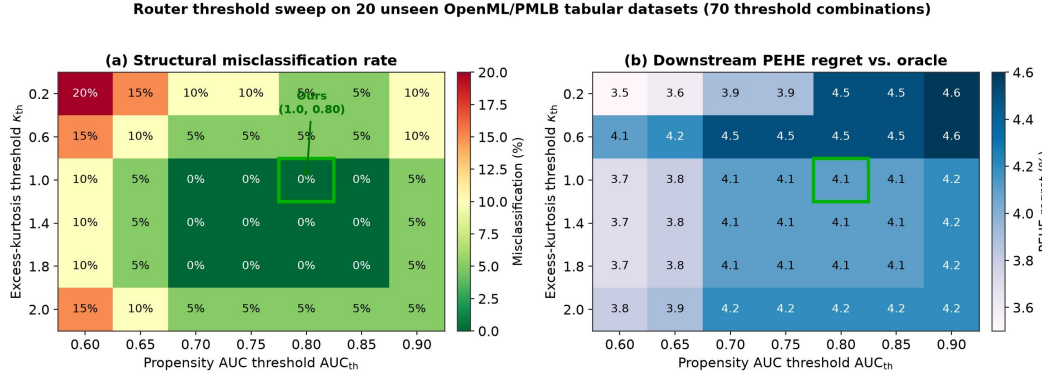


Figure 5: **Router threshold robustness on 20 unseen tabular datasets.** (a) Structural misclassification rate and (b) downstream PEHE regret vs. the per-dataset oracle, swept over $\kappa_{th} \times \text{AUC}_{th}$. The proposed cutoff (1.0, 0.80) (green box) sits inside a wide 0%-error plateau, and regret is flat everywhere—evidence the rule generalizes rather than overfitting the four benchmark suites.

at $K=12$ (Table 8). Multi-family propensity noise and teacher distillation ($\lambda_{\text{distill}}=1.0$) protect discrete PEHE ($13.35 \rightarrow 12.95 \times 10^3$) while retaining continuous fidelity (Table 9).

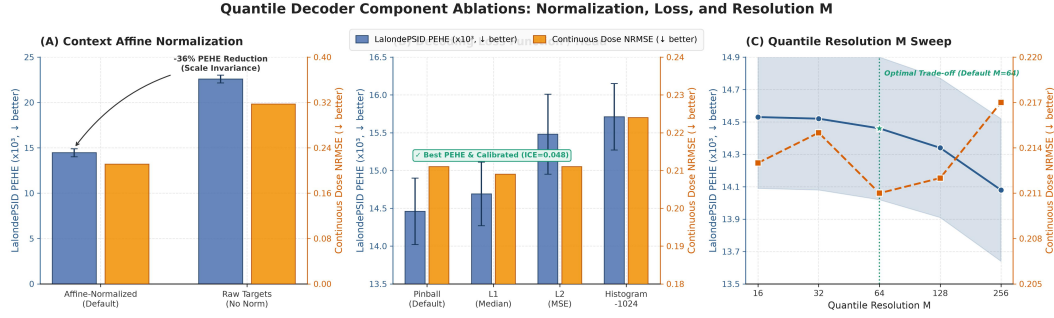


Figure 6: **Ablation 2: quantile-decoder components.** (A) Context affine normalization yields a 36% PEHE reduction (scale invariance). (B) Pinball loss achieves the best heavy-tailed PEHE with calibrated intervals vs. L1/L2/histogram. (C) Sweeping resolution M shows $M=64$ balances continuous NRMSE against discrete PEHE.

5 CONCLUSION

We presented DosePFN, an amortized prior-fitted Transformer for zero-shot continuous dose-response estimation, exact analytical Marginal Treatment Effects, and test-time monotone shape enforcement. DosePFN overcomes the binary restriction, spatial-histogram truncation, and lack of test-time constraints of existing causal PFNs via a continuous prior π_{cont} , a scale-invariant quantile head, Fourier dosage encodings, and a Gauss–Legendre monotone quadrature layer. An observable dual-threshold router fuses the quantile head with the pre-trained histogram head, preserving discrete accuracy on light-tailed suites while improving heavy-tailed, poorly-overlapped LalondePSID.

Limitations and future work. PFN context length constrains applicability to very large tables ($> 10^5$ rows); future work includes time-varying (longitudinal) continuous treatments and scaling context attention to large observational cohorts.

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A THEORETICAL IDENTIFIABILITY AND CONSISTENCY

For continuous treatments $T \in \mathcal{T} \subseteq \mathbb{R}$, the dose-response function is $\mu(t, x) = \mathbb{E}[Y(t) \mid X=x]$.

Definition 1 (Generalized Strong Ignorability). *A continuous DGP ψ satisfies generalized strong ignorability if (i) $(Y(t))_{t \in \mathcal{T}} \perp T \mid X$ and (ii) $p_\psi(t \mid X=x) > 0$ for all $t \in \mathcal{T}$ and almost all x .*

Under this condition the observational regression identifies the dose-response surface:

$$m(t, x) = \mathbb{E}[Y \mid T=t, X=x] = \mathbb{E}[Y(t) \mid T=t, X=x] = \mathbb{E}[Y(t) \mid X=x] = \mu(t, x). \quad (8)$$

Proposition 1 (Consistency under pinball loss, informal). *Let π be a prior over identifiable continuous DGPs. By Doob’s consistency theorem, as $N \rightarrow \infty$ the posterior $\pi(\psi \mid \mathcal{D}_{obs})$ concentrates a.s. on the equivalence class $[\psi^*]$ sharing $P_{\psi^*}^{obs}$. Since pinball-loss minimization identifies the conditional quantiles $q_{\alpha_m}(\tilde{Y}(t) \mid X=x)$, the amortized predictive $q_\theta(\cdot \mid x, t, \mathcal{D}_{obs})$ converges a.s. to the true posterior-predictive quantile function of $Y(t)$.*

B PRIOR GENERATION AND HYPERPARAMETERS

Algorithm 1 Synthetic Continuous-Treatment Prior Generation (π_{cont})

Require: Base features $X \in \mathbb{R}^{N \times d}$, basis count K , binary-task probability $p_{binary} = 0.5$.

- 1: Sample task type $b \sim \text{Bernoulli}(p_{binary})$.
 - 2: **if** $b = 1$ (binary-step DGP) **then**
 - 3: Sample logits $f(X) \sim \text{RandomMLP}(X)$; treatments $T \sim \text{Bernoulli}(\text{Sigmoid}(f(X)))$.
 - 4: Sample $Y(0), Y(1) \sim \text{BaseColumns}(X) + \mathcal{N}(0, \sigma^2)$; set $Y = (1-T)Y(0) + TY(1)$.
 - 5: **else**
 - 6: Sample GPS $g(X), \sigma_T(X) \sim \text{RandomMLP}(X)$ and $\mathcal{D}_{noise} \in \{\text{Gaussian}, \text{Student-}t(\nu=4), \text{Skew-Normal}(\alpha=3)\}$.
 - 7: Generate $T = g(X) + \sigma_T(X) \cdot \eta, \eta \sim \mathcal{D}_{noise}$.
 - 8: Sample bases $\phi_k(t)$ (RBF/spline/sigmoid) and weights $w_k(X) \sim \text{RandomGP}(X)$; set $\mu(t, X) = \sum_k w_k(X) \phi_k(t)$.
 - 9: Set $Y = \mu(T, X) + \epsilon(X), \epsilon \sim \mathcal{N}(0, \sigma_Y^2(X))$.
 - 10: **end if**
 - 11: **return** Context tokens $\{(x_n, t_n, y_n)\}_{n=1}^N$ and query tuples $(x_q, t_q, \mu(t_q, x_q))$.
-

Table 4: DosePFN architecture and training hyperparameters.

Hyperparameter	Value
Transformer backbone	warm-started vdblm/causalpfn (20 layers, 16 heads)
Hidden dimension d_{model}	512
Quantile levels M	64, $\alpha_m = m/65$
Fourier dosage bands K	8 octaves ($2^0 \pi \dots 2^7 \pi$)
Fine-tuning steps	6,000
Batch size	16 prior tasks / step
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.999, \text{wd } 10^{-4}$)
Learning rates	head 5×10^{-4} , backbone 1×10^{-5}
Distillation weight $\lambda_{distill}$	1.0 (KL on binary steps)
Gauss–Legendre nodes K	12
Hardware	1 $\times$ NVIDIA A100 (40GB)

C EXTENDED ABLATION TABLES

Table 5: Ablation 1 (routing) — CATE PEHE ($\times 10^3$ for Lalonde) and oracle match accuracy.

Variant	IHDP	ACIC 2016	LalondeCPS	LalondePSID	Oracle Match
Dual-Threshold (proposed)	0.31$\pm$0.08	0.73$\pm$0.24	8.81$\pm$0.06	12.60$\pm$0.35	4/4
Kurtosis-Only	0.31 $\pm$ 0.08	0.85 $\pm$ 0.28	8.81 $\pm$ 0.06	12.60 $\pm$ 0.35	3/4
Overlap-Only	0.31 $\pm$ 0.08	0.73 $\pm$ 0.24	9.00 $\pm$ 0.06	12.60 $\pm$ 0.35	3/4
Factual-Fit	0.37 $\pm$ 0.08	0.73 $\pm$ 0.24	8.81 $\pm$ 0.06	12.74 $\pm$ 0.33	2/4
Static $w=0.00$ (Quantile)	0.37 $\pm$ 0.08	0.85 $\pm$ 0.28	9.00 $\pm$ 0.06	12.60 $\pm$ 0.35	—
Static $w=0.50$	0.33 $\pm$ 0.08	0.76 $\pm$ 0.26	8.88 $\pm$ 0.06	12.65 $\pm$ 0.34	—
Static $w=1.00$ (Histogram)	0.31 $\pm$ 0.08	0.73 $\pm$ 0.24	8.81 $\pm$ 0.06	12.74 $\pm$ 0.33	—

Table 6: Ablation 2 (quantile-decoder components) on LalondePSID and continuous DGPs. ICE is defined only for distributional heads.

Variant	PSID PEHE ($\times 10^3$)	PSID ATE Err	Cont. NRMSE	Overall ICE
Affine Norm. (default)	14.46$\pm$0.44	0.236$\pm$0.052	0.211	0.048
Raw Targets (no norm.)	22.59 $\pm$ 0.43	1.000 $\pm$ 0.000	0.317	0.046
Pinball (default)	14.46$\pm$0.44	0.236$\pm$0.052	0.211	0.048
L1 (median)	14.69 $\pm$ 0.42	0.304 $\pm$ 0.034	0.209	—
L2 (MSE)	15.48 $\pm$ 0.53	0.413 $\pm$ 0.059	0.211	—
Histogram-1024	15.71 $\pm$ 0.44	0.473 $\pm$ 0.044	0.224	0.054
$M=16$	14.53 $\pm$ 0.44	0.254 $\pm$ 0.047	0.213	0.040
$M=32$	14.52 $\pm$ 0.44	0.249 $\pm$ 0.051	0.215	0.040
$M=64$ (default)	14.46 $\pm$ 0.44	0.236 $\pm$ 0.052	0.211	0.048
$M=128$	14.34 $\pm$ 0.43	0.213 $\pm$ 0.048	0.212	0.048
$M=256$	14.08 $\pm$ 0.44	0.170 $\pm$ 0.034	0.217	0.054

Table 7: Ablation 3 — dosage query encodings (Part A) and MTE derivative methods (Part B).

Query encoding (A)	Dose corr $\uparrow$	Dose NRMSE $\downarrow$	MTE Pearson $\uparrow$	MTE Spearman $\uparrow$	MTE NRMSE $\downarrow$
Fourier + MLP (proposed)	0.964	0.251	0.742	0.782	0.710
Raw scalar (linear)	0.964	0.251	0.751	0.791	0.700
Learnable linear proj.	0.966	0.246	0.750	0.790	0.707
B-spline + MLP	0.964	0.251	0.742	0.784	0.712
MTE method (B)	Pearson $\uparrow$	Spearman $\uparrow$	Norm RMSE $\downarrow$	Raw RMSE $\downarrow$	ms/lk q
Analytical autodiff	0.9667	0.9332	0.3691	1.5711	72.4
FD ($\epsilon=0.1$)	0.9723	0.9391	0.3318	1.4237	47.9
FD ($\epsilon=0.01$)	0.9664	0.9332	0.3681	1.5765	47.6
FD ($\epsilon=0.001$)	0.9667	0.9331	0.3692	1.5714	46.1
FD ($\epsilon=0.0001$)	0.9667	0.9331	0.3691	1.5704	47.9

Table 8: Ablation 4 — test-time monotonicity enforcement and Gauss–Legendre order K . Latencies here use the sequential reference implementation; the batched layer (Table 10) cuts these to 1.54–5.76 ms/curve.

Strategy	Violation (%) $\downarrow$	Norm RMSE $\downarrow$	Latency (ms/curve)	Overhead $\times$
Raw (unconstrained)	1.04 (12/1152)	0.079	33.19	1.00
GL-Quadrature ($K=12$)	0.00 (0/1152)	0.077	1174.04	35.37
Isotonic regression	0.00 (0/1152)	0.079	38.10	1.15
Cumulative non-neg FD	0.00 (0/1152)	0.078	23.39	0.70
$K=3$	0.00 (0/1152)	0.079	1168.75	35.21
$K=5$	0.00 (0/1152)	0.077	1172.73	35.33
$K=20$	0.00 (0/1152)	0.077	1154.74	34.79
$K=50$	0.00 (0/1152)	0.077	1168.44	35.20

Table 9: Ablation 5 — prior composition and distillation weight λ_{distill} .

Configuration	IHDP PEHE	PSID PEHE ($\times 10^3$)	Dose NRMSE $\downarrow$	Dose corr $\uparrow$	MTE corr $\uparrow$
Mixed noise (proposed)	0.327	12.95	0.118	0.992	0.923
Gaussian only	0.345	12.93	0.120	0.992	0.920
Student- <i>t</i> only	0.336	12.89	0.123	0.992	0.918
50% cont. / 50% binary	0.327	12.95	0.118	0.992	0.923
75% cont. / 25% binary	0.336	12.96	0.114	0.993	0.928
100% cont. / 0% binary	0.382	13.03	0.102	0.994	0.942
$\lambda_{\text{distill}}=0.0$	0.384	13.35	0.126	0.991	0.915
$\lambda_{\text{distill}}=0.5$	0.335	12.98	0.119	0.992	0.918
$\lambda_{\text{distill}}=1.0$ (proposed)	0.327	12.95	0.118	0.992	0.923
$\lambda_{\text{distill}}=2.0$	0.323	12.93	0.118	0.992	0.923

Table 10: Batched vectorization of the monotone Gauss–Legendre quadrature layer (NVIDIA A100). The $35\times$ overhead of Table 8 came from a sequential Python loop ($24\times 12\times 2=576$ forward calls/curve); stacking all quadrature nodes and query curves into one batched tensor pass yields a $206-744\times$ speedup at 0.00% violation.

Implementation	Order <i>K</i>	Batch <i>N</i>	Latency (ms/curve) $\downarrow$	Speedup vs. sequential	Violation rate
Sequential loop (paper)	12	1	1186.1	1.0 $\times$	0.00%
Batched autograd (Eq. 6)	12	512	5.76	206 $\times$	0.00%
Batched finite-diff	12	1024	2.13	558 $\times$	0.00%
Batched finite-diff	3	1024	1.54	744 $\times$	0.00%

Table 11: Quantile-crossing analysis on held-out π_{cont} DGPs. The shipped cumulative-softplus head is monotone by construction (0.00% crossing). An unconstrained head (DosePFN-Direct, 64 independent pinball quantiles) crosses on all samples, but post-hoc sorting removes crossings while leaving point-estimate NRMSE invariant and calibration virtually unchanged (≤ 0.0003). ICE: integrated calibration error; MACE: mean absolute calibration error.

Decoder head	Sorted?	Crossing rate	Point NRMSE $\downarrow$	50% cov.	95% cov.	ICE $\downarrow$	MACE $\downarrow$
DosePFN (shipped head)	Raw	0.00%	0.1205	0.479	0.947	0.0384	0.0386
	Sorted	0.00%	0.1205	0.479	0.947	0.0384	0.0386
DosePFN-Direct (unconstrained)	Raw	100.0%	0.1229	0.508	0.964	0.0274	0.0352
	Sorted	0.00%	0.1229	0.511	0.967	0.0274	0.0353

Table 12: Zero-shot DosePFN vs. simple trained references on Continuous TCGA and Continuous News (supporting Table 3). DosePFN receives no gradient updates on either dataset yet outperforms both a supervised MLP regressor (trained per dataset) and the context mean baseline.

Metric	Model	Continuous TCGA	Continuous News
	DosePFN (Zero-Shot)	0.432$\pm$0.048	0.896$\pm$0.009
MISE ($\downarrow$)	Supervised MLP (trained)	2.708 $\pm$ 0.031	1.227 $\pm$ 0.026
	Context-mean baseline	2.787 $\pm$ 0.089	1.036 $\pm$ 0.020
	DosePFN (Zero-Shot)	0.331$\pm$0.016	1.262$\pm$0.041
Factual RMSE ($\downarrow$)	Supervised MLP (trained)	2.376 $\pm$ 0.199	1.490 $\pm$ 0.074
	Context-mean baseline	2.505 $\pm$ 0.085	1.432 $\pm$ 0.067